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RR–R–1185
November 17, 1966
SUPERSEDING
(See 6.4)

FEDERAL SPECIFICATION

REFLECTOR AND FLAG WARNING KIT, HIGHWAY

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. The highway reflector and flag warning kit covered by this specification shall consist of three reflecting units with three red flags in a metal box, rack, or bracket (see 1.2). Each reflecting unit shall have two reflector elements (bidirectional) (see 3.3 and 3.4), and shall be a compact unit which can be quickly set up as a highway warning device. The reflecting units and red flags are used for warning the driver of an approaching vehicle of the presence of a stopped or disabled vehicle on the roadway.

1.2 Classification.

1.2.1 *Types.* Reflector kits shall be of the following types, as specified (6.2).

I—Reflecting units in a metal box, without provisions for vehicle mounting or mounting holes.

II—Reflecting units in a metal box, rack, or bracket, with provisions for vehicle mounting.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

UU–D–171—Decals.
UU–T–81—Tags, Shipping and Stock.
CCC–T–191—Textile Test Methods.
PPP–B–601—Boxes, Wood, Cleated-Plywood.
PPP–B–636—Box, Fiberboard.
PPP–B–640—Boxes, Fiberboard, Corrugated, Triple-Wall.

Federal Standards:

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 141—Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL–P–116—Preservation, Methods of.

Military Standards:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.
MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues in effect on date of invitation for bids or request for proposal shall apply.

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Nongovernmental:

Society of Automotive Engineers' (SAE) Standards:

Color Specification For Electric Lamps (SAE J578).

Emergency Reflex Reflectors (SAE J774).

Society of Automotive Engineers' Information Report:

Vibration Test Machine (SAE J577).

(Application for copies should be addressed to the Society of Automotive Engineers', Inc., 485 Lexington Ave., New York, N.Y., 10017.)

American Society for Testing and Materials (ASTM) Publications:

ASTM Designation: B 117—Standard for Salt Spray (Fog) Testing.

(Copies may be obtained from the American Society for Testing and Materials, 1916 Race St., Philadelphia, Pa., 19103.)

Color Association of the United States, Inc.:

Standard Color Card of America.

(Application for copies should be addressed to The Color Association of the United States, Inc., 200 Madison Ave., New York, N.Y., 10016.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Illustrations. The illustrations shown herein are for convenience of identification and are not intended to preclude the purchase of highway reflector and flag warning kits which are otherwise in accordance with the requirements of this specification.

3.2 Material. Material shall be as specified herein. Material not definitely specified shall be of good commercial quality for the purpose intended.

3.3 Reflecting units. Reflecting units shall consist of reflecting elements placed in a holder with a base. The reflecting elements shall be bidirectional and shall have equal reflectivity from the front and rear sides. The exposed surface of each face of each reflecting element shall be at least six square inches (minimum 2¾ inches in diameter). The reflecting unit holder shall be attached to, and supported by, a collapsible or folding base. The holder shall be securely and positively locked in the base, both when erected and when collapsed for storage. When erected, the holder shall be perpendicular to the base and the reflecting unit with flag in storage shall not tip or slide on a brushed concrete surface in a 40 mile per hour wind directed perpendicular to the front, back, or side as specified in 4.4.4. The weight of each reflecting unit shall be not more than 5 pounds. The height of the reflecting unit when erected, shall be not less than 7 nor more than 12 inches. Each reflecting unit shall be furnished with a socket, loop, or similar means for securely holding the flag and flagstaff. When the flagstaff is in place on the reflecting unit, the flag shall not cover any portion of the reflecting surface and shall hang off center.

3.3.1 *Reflecting element.* The reflectors shall be either reflex-reflector sheeting or reflex-reflector elements. The reflex-reflector sheeting shall have the reflective face placed behind and sealed to red methyl methacrylate plastic material. The reflex-reflective elements shall be made of red methyl methacrylate plastic. The red color shall be in accordance with the SAE Standard Color Specification for Electric Lamps (see 4.4.5). The reflector element shall be composed of one completely sealed bidirectional reflector piece. As an alternative, the reflector element may be composed of two separate reflector pieces, each piece being of a single direction, which when inserted in its position in the holder shall be placed back to back. When the reflector element is composed of two separate reflector pieces, each piece shall be individually sealed. Sealing of all reflecting elements shall be accomplished to prevent moisture or dirt from reaching the reflecting surfaces. The seal shall pass the test specified in 4.4.1. The two reflector elements together for each side shall have a minimum effective return candlepower of not less than those specified in table I, when tested as specified in 4.4.2.

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TABLE I. *Minimum candlepower per incident foot-candle*

Observation angle	Entrance				
	0°	10° up	10° down	20° left	20° right
Degrees					
0.2	18.0	12.0	12.0	5.0	5.0
1.5	0.14	0.10	0.10	0.06	0.06

3.4 Reflector elements. Each reflecting unit shall consist of two reflecting elements located one above the other (see fig. 1).

3.4.1 *Flags.* The flags furnished with reflecting units shall be at least 12 by 12 inches square. The flags shall be fabricated of cotton sheeting and shall be of a color corresponding to cable number 70180 (Old Glory Red) of the Standard Color Card of America. The material shall have fair or better colorfastness to water and light when tested as specified in 4.4.3. The flag edges shall be bound or rolled and stitched to prevent ravelling.

3.4.2 *Flagstaffs.* The flagstaffs shall be of such length that the top of the flag shall be not less than 17 inches from the surface the unit stands upon. The flagstaff, for compactness, shall be capable of disassembly or folding so that it will fit the box, rack, or bracket as specified herein.

3.4.3 *Kit.*

3.4.3.1 *Type I.* The highway reflector kit shall consist of a metal box containing 3, two element, bidirectional reflecting units, three flags and flagstaffs, and instructions.

3.4.3.2 *Type II.* The highway reflector kit shall consist of a metal box, rack, or bracket, as applicable, with provisions for vehicle mounting, containing 3, two element, bidirectional reflecting units, three flags and flagstaffs, and instructions.

3.5 Instructions. Each highway reflector kit shall be furnished with legible instructions enclosed in the case and adequately weatherproofed. When specified (see 6.2), instructions shall be furnished on a tag conforming to UU–T–81, type A or B, class 1 or 2, or on a decal conforming to UU–D–171. The instructions shall detail the procedure required for the erection of the reflecting units and include the following instructions as to placement of the units for use:

INSTRUCTIONS FOR USE

PLACE ONE REFLECTOR 40 PACES (100 FEET) BEHIND AND ONE REFLECTOR 40 PACES AHEAD OF THE STOPPED VEHICLE.

PLACE ONE REFLECTOR 10 FEET FROM THE VEHICLE TOWARD THE APPROACHING TRAFFIC.

If the vehicle is blocking a traffic lane, all reflectors shall be placed in that traffic lane.

If the vehicle is on the shoulder of the roadway, reflectors shall be placed at the edge of the roadway.

If the vehicle is stopped near a hill or curve or obstruction in the road, reflectors shall be placed at least 40 paces, but not more than 120 paces to the front and rear.

In daylight hours, flags shall be attached to the reflectors farthest from the vehicle.

3.6 Vibration resistance. The highway reflector kits shall show no evidence of structural failure or damage reflecting elements at the completion of the vibration test specified in 4.4.6.

3.7 Finishing and painting. The finish shall withstand the corrosion-resistance test in 4.4.7. Unless otherwise specified (see 6.2), the color of the equipment, except the reflecting elements and flags, shall be the manufacturer's standard.

3.8 Identification marking. The equipment shall be marked in a plain and permanent manner with the name of the manufacturer or trademark of such known character that the source of manufacture can be readily identified.

3.9 Workmanship. Workmanship shall be in accordance with good commercial practice of this type of equipment. Threads, if used, shall be of the national form. Rivets, if used, shall be firmly and neatly set. All burrs and sharp

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edges which might cause injury to the operator shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for inspection. Sampling for inspection shall be performed in accordance with the provisions set forth in MIL–STD–105, except where otherwise indicated. For purposes of sampling, an inspection lot for examination and tests shall consist of all material of the same type and class submitted for delivery at one time.

4.2.1 *Inspection of components and materials.* In accordance with 4.1 the supplier is responsible for insuring that components and materials used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or, if none, in accordance with this specification. A certificate of compliance from the contractor, or his supplier, certifying that the cotton sheeting used in the fabrication of the flags shall have fair or better colorfastness to water and light when tested as specified in 4.4.3, shall satisfy the requirement for this characteristic requirement.

4.2.1.1 *Testing of components.* In addition to the quality assurance provisions of the subsidiary specifications and drawings, colorfastness tests shall be performed on the flag for compliance with the requirement in 3.4.1 and 4.4.3. If the dyeing operation is performed by contractor's supplier, these test results furnished by the supplier will be acceptable. The sample unit for this test shall be one flag and the sample size shall be five. Failure of any one sample unit to meet colorfastness requirements shall be cause for rejection of the lot.

4.3 Inspection of the end item.

4.3.1 *Examination of the end item.* Examination of the end item shall be in accordance with the classification of defects set forth below. The lot size, for purpose of determining the sample size, shall be expressed in units of reflector kits for examinations in 4.3.1.1 and 4.3.1.2. For examination in 4.3.1.3, the lot size shall be expressed in units of completely packaged, packed, sealed, and marked shipping containers. The inspection levels, for determining the sample size, and the acceptable quality levels (AQL's) shall be as set forth in 4.3.1.4.

4.3.1.1 *Examination of the end item for defects in construction, appearance, and workmanship.* The sample unit for this examination shall be one complete reflector kit.

Examine	Defect	Major	Minor
Construction and appearance.	Not type specified	X	
	Holder not securely supported by or perpendicular to base	X	
	Frame for reflecting element does not operate freely and uniformly		X
	Reflecting element not bidirectional	X	
	Reflecting element loose		X
	Reflecting element not sealed	X	
	Reflecting element cracked, crazed, or chipped	X	
	Parts or components incorrect, missing or damaged	X	
	Any component out of alignment in a manner that interferes with easy and complete assembly and use	X	
	Metal box, rack, or bracket does not provide protection against damage to reflecting unit	X	

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Examine	Defect	Major	Minor
	Finish not uniform in color; any area lacking finish; any scratch or abrasion in finish		X
	Flag not made of cotton sheeting	X	
	Flag not uniform in color; does not visually match color specified		X
	Flag edge not bound or rolled to prevent ravelling		X
	Flagstaff not capable of dissassembly or folding for compactness	X	
	Flagstaff does not properly and securely fit socket		X
	Any portion of flag, covering part of reflecting unit when in place		X
	Instruction incorrect, illegible, or missing		X
	Name or trademark of manufacturer missing or illegible		X

4.3.1.2 *Examination of the end item for dimensional-weight defects.* The sample unit for this examination shall be one reflector kit.

Examine	Defect
Reflecting unit	Less than 7 or more than 12 inches in height.
Reflecting element	Less than 2¾ inches in diameter.
Flag	Less than 12 by 12 inches.
	Top of flag when mounted less than 17 inches from the surface of stand.
Weight of each reflecting unit.	More than 5 pounds.

4.3.1.3 *Examination of preparation for delivery.* The sample unit for this examination shall be one completely packaged, packed, sealed, and marked shipping container.

Examine	Defect
Preservation and packaging (levels A and B)	Unprotected metal surfaces subject to corrosion not cleaned, dried, and coated with preservative, as specified.
	Complete reflector kit not packaged in fiberboard box, as specified.
	Reflector kit not adequately cushioned for protection against movement or impact.
	Fiberboard unit container not closed and sealed; joints and seams not sealed with gummed tape or pressure sensitive tape; tape does not extend over all corners.
(Level C)	Not preserved and packaged to afford protection against deterioration and damage.
Packing	Not type shipping container specified, as applicable.
Workmanship	Inadequate application of components, such as insufficient or incomplete closure of container flaps, case liner, loose strapping, bulge or distortion of container; container bruised, broken, or damaged in a manner which could affect serviceability or cause loss or damage to content.
Contents	Number and arrangement of packages not as specified or indicated.
Marking (interior and exterior)	Omitted, incomplete, incorrect, illegible; of improper size, location, sequence or method of application.
Weight (gross)	Exceeds maximum requirements.

4.3.1.4 *Inspection levels and AQL's for examinations.* The inspection levels for determining the sample size, and the AQL's expressed in defects per hundred units, shall be as follows:

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Examination paragraph	Inspection level	AQL	
		Major	Total
4.3.1.1	I	4.0	10.0
4.3.1.2	S–1	---	4.0
4.3.1.3	S–2	---	4.0

4.3.2 *Testing of the end item.* The end item shall be tested for the applicable characteristics indicated in table II, herein, on samples selected from each lot presented for examination. The lot size shall be expressed in units of reflector kits. The inspection levels and AQLs shall be as indicated in table II. The sample unit shall be two complete reflector kits. These sample units, randomly selected throughout the lot, shall be tested for each characteristic with no evidence of failure to meet the specified quality levels.

4.4 **Tests.** In lieu of any or all of the following tests, at the option of the contracting officer, evidence may be accepted that the reflectors have been tested in accordance with this specification by a testing laboratory acceptable to the procuring activity.

4.4.1 *Reflecting element seal test.* The reflecting elements shall be submerged in water at room temperature in a chamber and the chamber evacuated to an absolute pressure of not more than 5 inches of mercury and held at this pressure for not less than 5 minutes. At the end of the 5-minute period, the vacuum shall be released, and the samples allowed to remain submerged for not less than 5 minutes and then removed from the water and visually examined for evidence of entrance of water. The presence of water shall be cause for rejection of the kit (see 3.3.1).

4.4.2 *Candlepower test.* The reflecting elements shall be tested in accordance with SAE J774. Candlepower values shall be not less than those specified in table I (see 3.3). Reflectors having values lower than those specified shall result in the rejection of the kit.

4.4.3 *Colorfastness tests.* One flag shall be taken from the sample kit and tested in accordance with CCC–T–191, method 5630, for colorfastness to water, and method 5660 or 5662, for colorfastness to light. Failure of the sample to have "fair" or "better" colorfastness shall be cause for rejection (see 3.4.1).

TABLE II. *Instructions for testing (sample unit)*

Characteristic	Specification reference		Requirements applicable to		Number determinations per sample unit	Results reported as		Inspection level	AQL
	Requirement	Test method	Individual unit	Lot average		Pass or fail [1]	Numerically to nearest [2]		
									(Expressed in defects per 100 units)
Wind test	3.3	4.4.4	X	---	1	X	---	S–2	2.5
Color test (reflecting elements)	3.3.1	4.4.5	X	---	1	X	---	S–2	2.5
Reflecting element seal test	3.3.1	4.4.1	X	---	1	X	---	S–2	2.5
Candle power test	3.3.1 & table I	4.4.2	X	---	1	---	0.01	S–2	2.5
Vibration	3.6	4.4.6	X	---	1	X	---	S–2	2.5
Corrosion-resistance	3.7	4.4.7	X	---	1	X	---	S–2	2.5

[1] If failure is indicated, report description of failure.
[2] Test reports shall include all values on which results are based, where applicable.

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4.4.4 *Wind test.* The ability to withstand wind pressure as specified in 3.3, shall be tested as follows: The reflecting unit, with flag in storage, shall be set upon a horizontal brushed concrete or other similar surface and subjected to a wind of 40 m.p.h. horizontal velocity, striking the unit perpendicular to the reflecting faces, with the wind directed first against one face, then the other and then four intermediate positions. Tipping over or sliding on the supporting surface shall each be cause for rejection.

4.4.5 *Color test.* The reflecting elements shall be checked for color in accordance with SAE J–578. Failure of the elements to conform to the standard is cause for rejection of all kits in the lot (see 3.3.1).

4.4.6 *Vibration test.* The highway reflector kits without flags or staff shall be subjected to a vibration test using a machine similar to figure 2 (see 3.6). Three reflecting units in the metal box, rack, or bracket in which they were submitted for test, shall be bolted to the anvil end of the table of the vibration-test machine and vibrated approximately 750 times per minute through a distance of ⅛ inch. The table shall be spring-mounted at one end and fitted with steel calks on the underside of the other end. These calks are to make contact with the steel anvil once during each cycle at the completion of the fall. The rack shall be operated under a spring tension of 60 to 70 pounds. This test shall be continued for one hour.

Note: The vibration test machine illustrated in figure 2 is the same as the machine covered by SAE–J577.

4.4.6.1 At the completion of the vibration test, the reflector kits shall be examined. Any reflector kit showing evidence of structural failure or damaged reflector elements at any time during this test, fails to meet the requirements of this specification.

4.4.7 *Corrosion-resistance test.* A sample unit shall be subjected to a salt spray (fog) test in accordance with either the latest ASTM B 117 or Fed. Test Method Std. No. 141, (method 6061). The sample unit shall be exposed to a 5 percent (NaCl) spray for a period of 50 hours, consisting of two periods of 24 hours' exposure and one hour minimum drying time each. There shall be no evidence of excessive corrosion immediately after the test has been completed, which would affect the proper functioning of the reflector kit.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 *Level A.* Each reflector kit shall be preserved and packaged in accordance with method III of MIL–P–116, utilizing a close-fitting box conforming to PPP–B–636, W6c, as the unit container. The box shall be closed in accordance with the appendix to the box specification.

5.1.2 *Level B.* Each reflector kit shall be packaged in a close-fitting box conforming to PPP–B–636, class domestic. Cushioning shall be provided. The box shall be closed in accordance with the appendix to the box specification.

5.1.3 *Level C.* The reflector kits shall be preserved and packaged to afford adequate protection against corrosion, deterioration, and damage during shipment from the supply source to the first receiving activity. The supplier's commercial practice will be acceptable when it fulfills the above requirements.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.* Reflector kits of like description, packaged as specified in 5.1, shall be packed in a close-fitting box conforming to PPP–B–601, overseas type. The box shall be closed and strapped in accordance with the appendix to the box specification. The gross weight shall not exceed 200 pounds.

5.2.2 *Level B.*

5.2.2.1 *Military agencies.* Reflector kits, packaged as specified in 5.1, shall be packed in a close-fitting box conforming to PPP–B–636, V3c or V3s; or PPP–B–640, grade A, class 2. The box shall be closed in accordance with the appendix to the box specification. The gross weight of the PPP–B–636 box shall not exceed 90 pounds; the gross weight of the triple-wall box shall not exceed 200 pounds.

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5.2.2.2 *Civil agencies.* Unit packages of reflector kits of like description in quantities as specified (see 6.2), shall be packed in a close-fitting box conforming to PPP–B–636, class domestic. The box shall be closed in accordance with the appendix to the box specification.

5.2.3 *Level C.* The reflector kits shall be packed to insure carrier acceptance and safe delivery at destination at the lowest rates, in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military agencies.* In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL–STD–129.

6. NOTES

6.1 Intended use. Each reflector and flag warning kit is intended for use on roadways by drivers of stopped vehicles as a warning to approaching vehicles. The reflecting units only, are to be used for night, and the reflecting units and flags for daylight hours.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(*a*) Title, number, and date of this specification.
(*b*) Type required (see 1.2.1).
(*c*) Instructions on tag or decal, when required (see 3.5).
(*d*) Color of the equipment, if different (see 3.7).
(*e*) Quantities of unit packages of reflector and flag warning kits required in shipping container (see 5.2.2.2).
(*f*) Selection of applicable levels of preservation, packaging, and packing required (see 5.1 and 5.2).

6.3 Military agencies. Level B preservation, packaging, and packing is intended to provide economical but limited protection and should be specified only when it is determined the reflector and flag warning kits will be held in covered storage.

6.4 Supersession data. This specification supersedes RR–F–440a, Flare Kit, Reflector Type, With Flags, dated October 8, 1957; and RR–F–00440c (GSA–FSS) Reflector Kit, Highway Warning, with Flags, dated December 24, 1964.

Comparison table

RR–R–1185	RR–F–00440	RR–F–440
None	None	Class 1, style A.
Type I	Type I	Class 1, style B.
None	None	Class 1, style A.
Type II	Type II	Class 1, style B.

MILITARY CUSTODIANS:

Army—MO(ERDL)
Navy—YD
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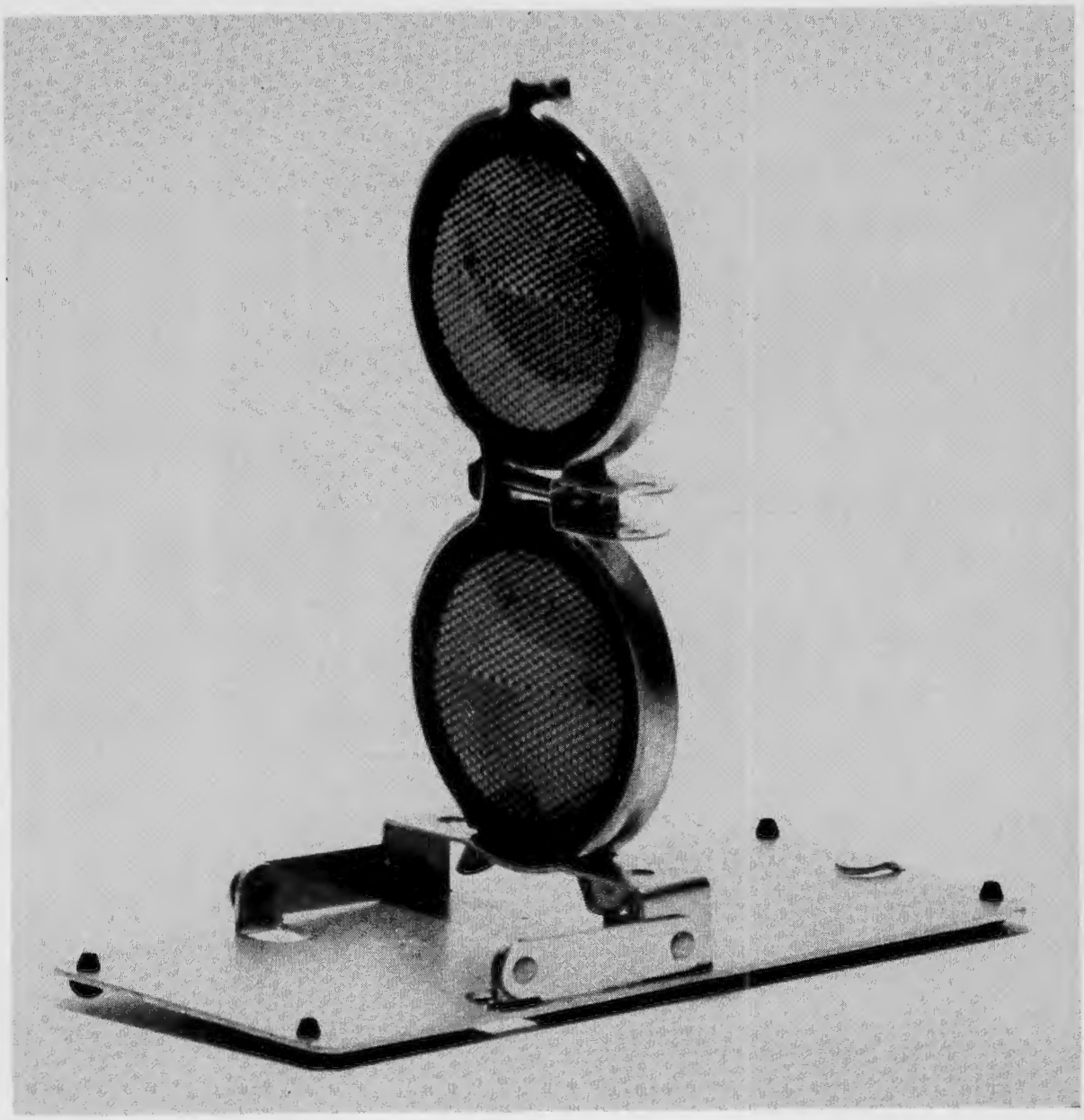


FIGURE 1. Reflecting unit shown in set-up position.

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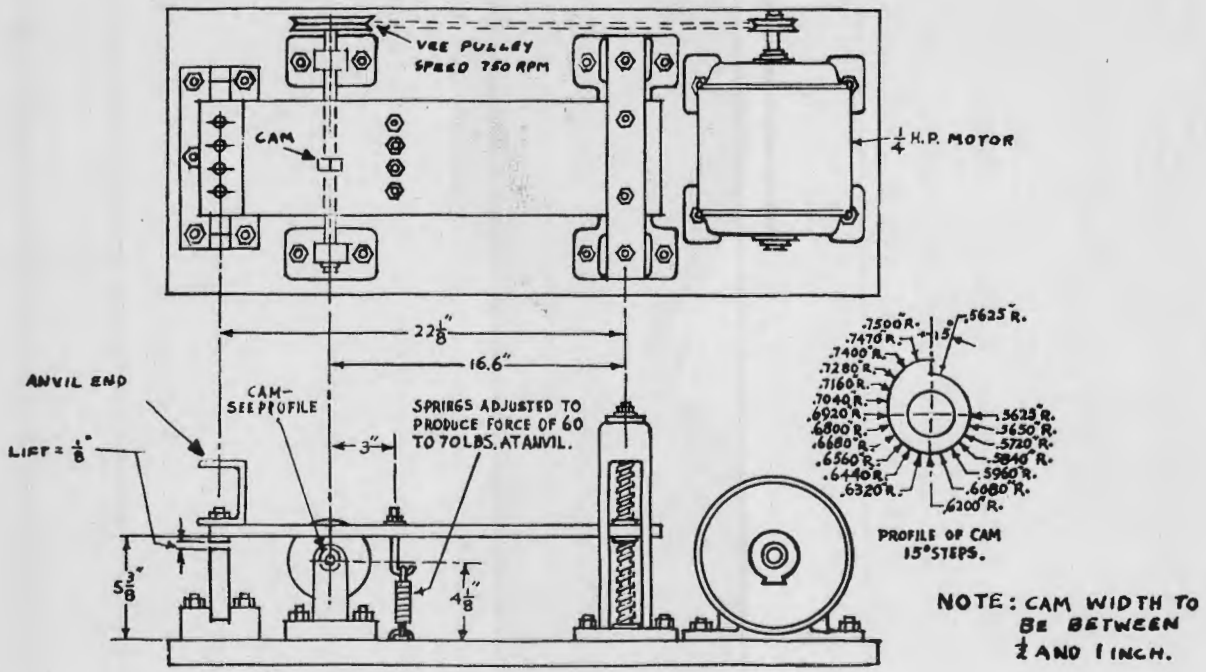


FIGURE 2. Vibration test machine.

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price cents each.

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